

Supermarket Sales Analysis

Project Report

IBM SkillsBuild | Data Analytics Project | 2024

Author: Chettukadi Tharun | Platform: Google Colab | Language: Python 3

Dataset: data/supermarket_sales.csv | Date Range: 2024-01-02 to 2024-12-31 | Rows: 500 | Columns: 13

1. Problem Statement

Retail supermarkets generate large volumes of transaction data daily but often lack structured analytical frameworks to extract actionable business intelligence from this data. The objective of this project is to analyse 500 supermarket sales transactions recorded across three branches in Delhi, Bangalore, and Mumbai during January–December 2024, and to derive data-driven insights that support product management, customer engagement, branch operations, payment strategy, and revenue growth decisions.

2. Dataset

File: data/supermarket_sales.csv | Rows: 500 | Columns: 13 | Period: 2024-01-02 to 2024-12-31

The dataset contains 500 unique supermarket transactions with the following fields: Invoice ID, Date, Branch, City, Customer Type, Gender, Product, Category, Quantity, Unit Price, Payment Method, Customer Rating, and Sales (= Quantity × Unit Price). Three branches: Branch A (Delhi), Branch B (Bangalore), Branch C (Mumbai). The dataset covers 15 products across 6 categories: Beverages, Dairy, Grains, Snacks, Bakery, and Grocery.

Column	Type	Description
Invoice ID	String	Unique transaction identifier
Date	Date (YYYY-MM-DD)	Transaction date — range 2024-01-02 to 2024-12-31
Branch	String	Branch A / B / C
City	String	Delhi / Bangalore / Mumbai
Customer Type	String	Member / Normal
Gender	String	Female / Male
Product	String	15 distinct products
Category	String	6 categories
Quantity	Integer	Units purchased (1–6)
Unit Price	Float (Rs)	Price per unit (Rs 45–Rs 200)
Payment	String	UPI / Cash / Credit Card
Rating	Float	3.5 – 4.8
Sales	Float (Rs)	Quantity × Unit Price

3. Data Cleaning

Data quality checks performed on the full 500-row dataset:

- Missing values: 0 across all 13 columns
- Duplicate rows: 0 (all Invoice IDs are unique)
- Date column: converted from string to datetime format; Month and Month_Num columns derived

- Sales formula validation: Sales = Quantity × Unit Price confirmed across all rows
- Categorical consistency: Branch, City, Payment, Customer Type, Gender all contain expected values only
- Result: Dataset is complete, consistent, and ready for analysis with no records removed.

4. Exploratory Data Analysis

Key descriptive statistics: Sales range Rs 225–Rs 1,200 per transaction; average Rs 488.82. Quantity per transaction ranges 1–6 units; average 5.54. Customer ratings range 3.5–4.8; average 3.99. Unit prices range Rs 45 (Biscuits) to Rs 200 (Paneer). The dataset spans all 12 months of January–December 2024: January 41 transactions, all other months 42 (except December 39). Gender split: 51.4% Female (257), 48.6% Male (243). Customer Type: 52.8% Member (264), 47.2% Normal (236). Payment methods: UPI 38.8% (194), Cash 29.6% (148), Credit Card 31.6% (158). Cashless (UPI + Credit Card): 70.4% of transactions.

5. KPI Analysis

KPI	Value	Source
Total Sales	Rs 244,411.08	sum(Sales)
Total Transactions	500	row count
Average Transaction Value (ATV)	Rs 488.82	Total Sales / Transactions
Average Customer Rating	3.99 / 5	mean(Rating)
Total Quantity Sold	2,768 units	sum(Quantity)
Top Product (by revenue)	Cheese	Rs 58,066.08
Top Category (by revenue)	Beverages	Rs 117,511.08
Top Branch (by revenue)	Branch C	Rs 87,956.58
Top City (by revenue)	Mumbai	Rs 87,956.58
Most Used Payment Method	UPI	194 txn (38.8%)
Peak Revenue Month	April 2024	Rs 22,495.00
Lowest Revenue Month	December 2024	Rs 17,361.50

All KPI values calculated directly from data/supermarket_sales.csv.

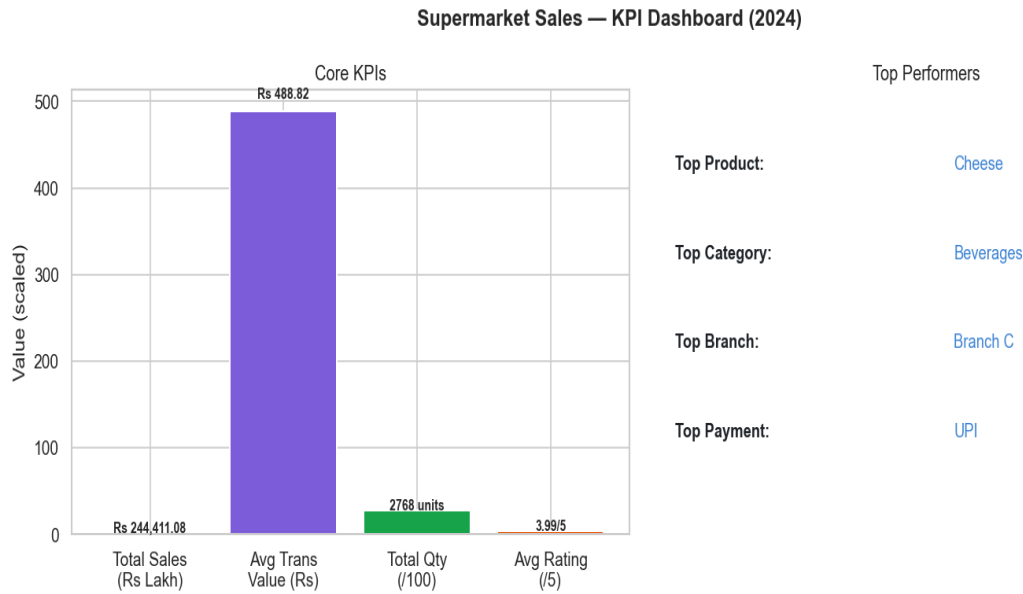


Figure 1: KPI Dashboard — computed from data/supermarket_sales.csv

6. Product Analysis

The dataset contains 15 products across 6 categories. Cheese (Rs 185.50/unit) is the top product by total sales revenue — Rs 58,066.08 across 55 transactions (11.0% of all transactions). Paneer (Rs 200/unit) has the highest unit price; individual Paneer transactions involving 4–6 units generate per-invoice values up to Rs 1,200. Tea (62 transactions) and Rice (58 transactions) are the two highest-frequency products by transaction count.

Product	Unit Price (Rs)	Transactions	Units Sold	Total Sales (Rs)
Cheese	Rs 186.26	55	313	Rs 58,066.08
Rice	Rs 80.00	58	348	Rs 27,840.00
Tea	Rs 75.00	62	359	Rs 26,925.00
Milk	Rs 95.00	32	192	Rs 18,240.00
Eggs	Rs 65.00	54	270	Rs 17,550.00
Yogurt	Rs 85.00	28	168	Rs 14,280.00
Bread	Rs 55.00	40	200	Rs 11,000.00
Wheat Flour	Rs 90.00	20	120	Rs 10,800.00
Chips	Rs 55.00	36	180	Rs 9,900.00
Juice	Rs 110.00	14	84	Rs 9,240.00
Biscuits	Rs 45.00	38	190	Rs 8,550.00
Sugar	Rs 50.00	34	170	Rs 8,500.00
Oats	Rs 120.00	10	60	Rs 7,200.00
Butter	Rs 120.00	9	54	Rs 6,480.00
Coffee	Rs 140.00	6	36	Rs 5,040.00

Paneer	Rs 200.00	4	24	Rs 4,800.00
--------	-----------	---	----	-------------

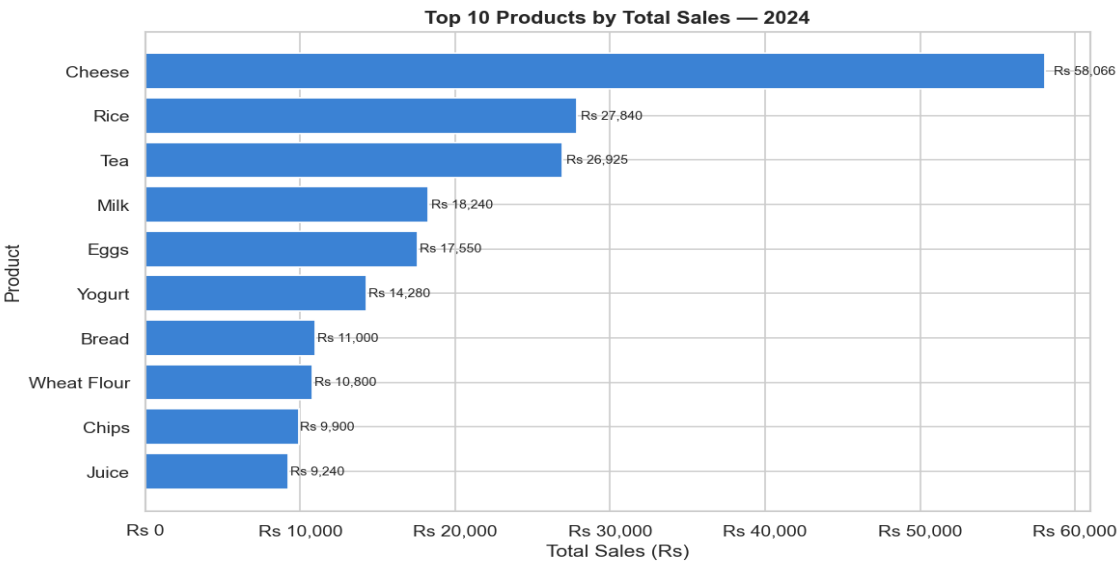


Figure 2: Top Products by Total Sales (2024)

7. Category Analysis

Beverages is the top category — 169 transactions (33.8% of all), Rs 117,511.08 revenue (48.1% of total). It spans both premium products (Cheese at Rs 185.50, Coffee at Rs 140) and everyday necessities (Milk at Rs 95, Tea at Rs 75). Grains (Rs 45,840.00) and Dairy (Rs 43,110.00) are the second and third categories. Bakery and Grocery are single-product categories (Bread and Sugar) representing the smallest revenue contributors.

Category	Transactions	% of Total	Total Sales (Rs)	% Revenue	Avg Rating
Beverages	169	33.8%	Rs 117,511.08	48.1%	3.986
Grains	88	17.6%	Rs 45,840.00	18.8%	3.966
Dairy	95	19.0%	Rs 43,110.00	17.6%	4.018
Snacks	74	14.8%	Rs 18,450.00	7.5%	4.011
Bakery	40	8.0%	Rs 11,000.00	4.5%	3.993
Grocery	34	6.8%	Rs 8,500.00	3.5%	3.947

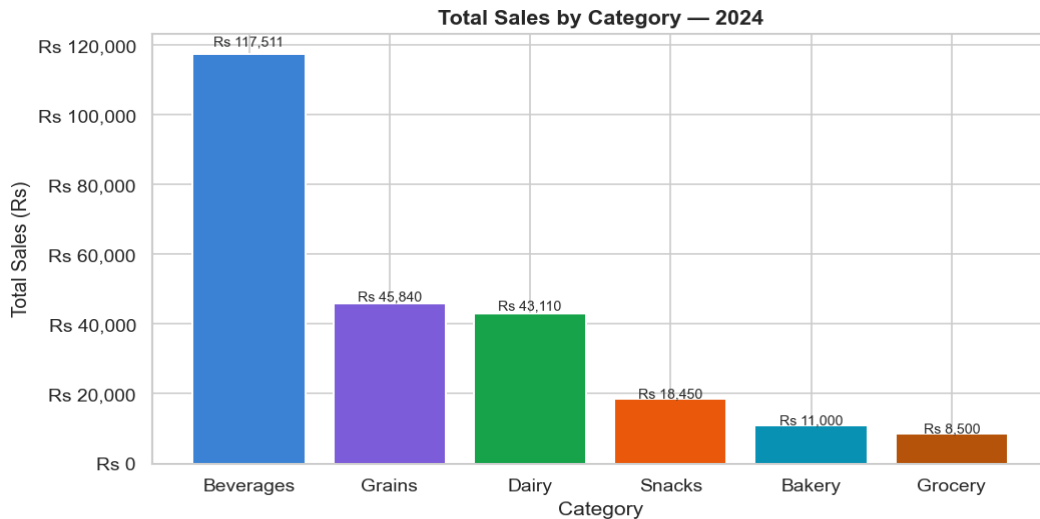


Figure 3: Total Sales by Category (2024)

8. Branch Analysis

Branch C (Mumbai) is the top-performing branch: Rs 87,956.58 (36.0%, 180 transactions). Branch B (Bangalore): Rs 78,555.00 (32.1%, 170 transactions). Branch A (Delhi): Rs 77,899.50 (31.9%, 150 transactions). All three branches contribute approximately one-third each of total annual revenue — a healthy operational balance.

Branch	City	Transactions	Total Sales (Rs)	% Revenue	Avg Rating
Branch C	Mumbai	180	Rs 87,956.58	36.0%	3.825
Branch B	Bangalore	170	Rs 78,555.00	32.1%	4.104
Branch A	Delhi	150	Rs 77,899.50	31.9%	4.059

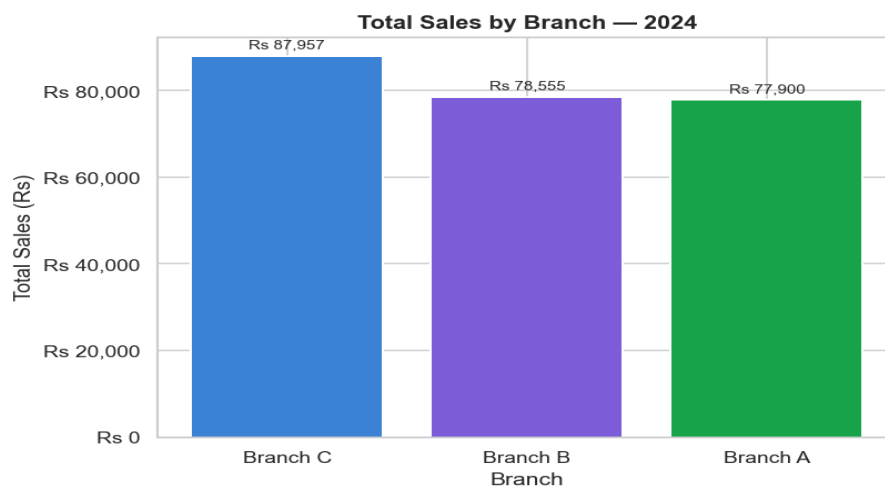


Figure 4: Total Sales by Branch (2024)

9. City Analysis

Mumbai (Branch C) is the top-performing city: Rs 87,956.58 (180 transactions). Bangalore: Rs 78,555.00 (170 transactions). Delhi: Rs 77,899.50 (150 transactions). The near-equal three-city revenue distribution

indicates the business is not over-dependent on any single geographic market.

City	Branch	Transactions	Total Sales (Rs)	% Revenue
Mumbai	Branch C	180	Rs 87,956.58	36.0%
Bangalore	Branch B	170	Rs 78,555.00	32.1%
Delhi	Branch A	150	Rs 77,899.50	31.9%

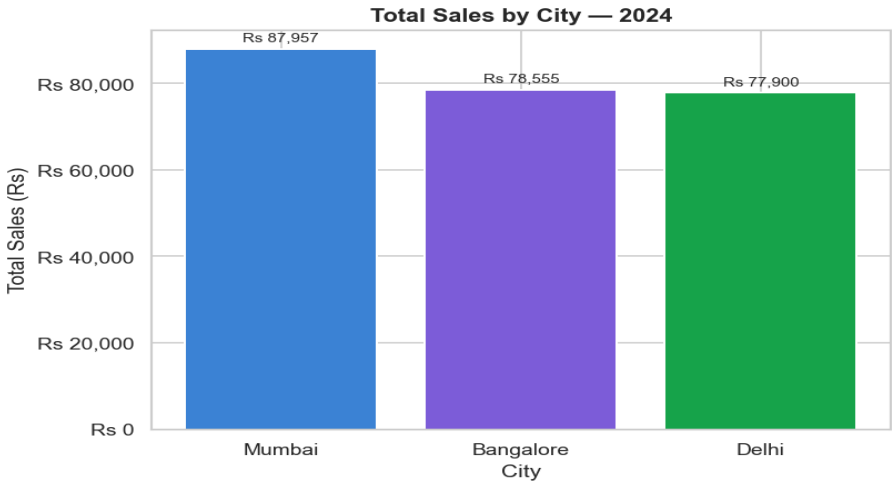


Figure 5: Total Sales by City (2024)

10. Payment Analysis

UPI is the most used payment method: 194 transactions (38.8%). Cash: 148 transactions (29.6%). Credit Card: 158 transactions (31.6%). Cashless (UPI + Credit Card): 352 transactions (70.4%). This reflects India's rapid digital payments adoption across urban markets.

Payment Method	Transactions	% of Total
UPI	194	38.8%
Credit Card	158	31.6%
Cash	148	29.6%
Cashless Total (UPI + Credit Card)	352	70.4%

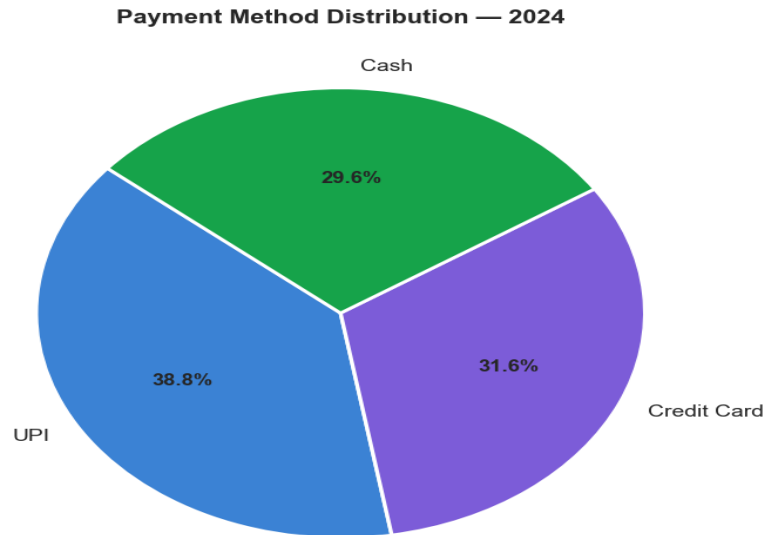


Figure 6: Payment Method Distribution (2024)

11. Customer Type Analysis

Member customers: 264 transactions (52.8%), Rs 127,596.08 revenue, ATV Rs 483.32, Avg Rating 3.978.
Normal customers: 236 transactions (47.2%), Rs 116,815.00 revenue, ATV Rs 494.98, Avg Rating 4.003.
Normal customers show a marginally higher ATV. Both segments are closely matched. The Normal customer base is the primary conversion opportunity for growing Member share.

Customer Type	Transactions	% of Total	Total Sales (Rs)	ATV (Rs)	Avg Rating
Member	264	52.8%	Rs 127,596.08	Rs 483.32	3.978
Normal	236	47.2%	Rs 116,815.00	Rs 494.98	4.003



Figure 7: Member vs Normal Customer Comparison (2024)

12. Monthly Sales Trend

The dataset covers all 12 months of January–December 2024. All figures calculated directly from data/supermarket_sales.csv. Peak revenue month: April 2024 — Rs 22,495.00 (42 transactions). Lowest revenue month: December 2024 — Rs 17,361.50 (39 transactions). H1 (Jan–Jun): Rs 124,706.08 — 51.0% of revenue (251 transactions). H2 (Jul–Dec): Rs 119,705.00 — 49.0% (249 transactions). H1 and H2 are nearly balanced. Q4 (58,409.00 — 23.9%) is the weakest quarter.

Month	Year	Transactions	Total Sales (Rs)	% Annual Sales
January	2024	41	Rs 21,547.00	8.8%
February	2024	42	Rs 18,938.50	7.7%
March	2024	42	Rs 22,303.08	9.1%
April	2024	42	Rs 22,495.00	9.2%
May	2024	42	Rs 19,025.50	7.8%
June	2024	42	Rs 20,397.00	8.3%
July	2024	42	Rs 21,951.00	9.0%
August	2024	42	Rs 20,354.00	8.3%
September	2024	42	Rs 18,991.00	7.8%
October	2024	42	Rs 20,208.50	8.3%
November	2024	42	Rs 20,839.00	8.5%
December	2024	39	Rs 17,361.50	7.1%
TOTAL	2024	500	Rs 244,411.08	100.0%

Quarter	Months	Transactions	Total Sales (Rs)	% Annual Sales
Q1	Jan–Mar	125	Rs 62,788.58	25.7%
Q2	Apr–Jun	126	Rs 61,917.50	25.3%
Q3	Jul–Sep	126	Rs 61,296.00	25.1%
Q4	Oct–Dec	123	Rs 58,409.00	23.9%

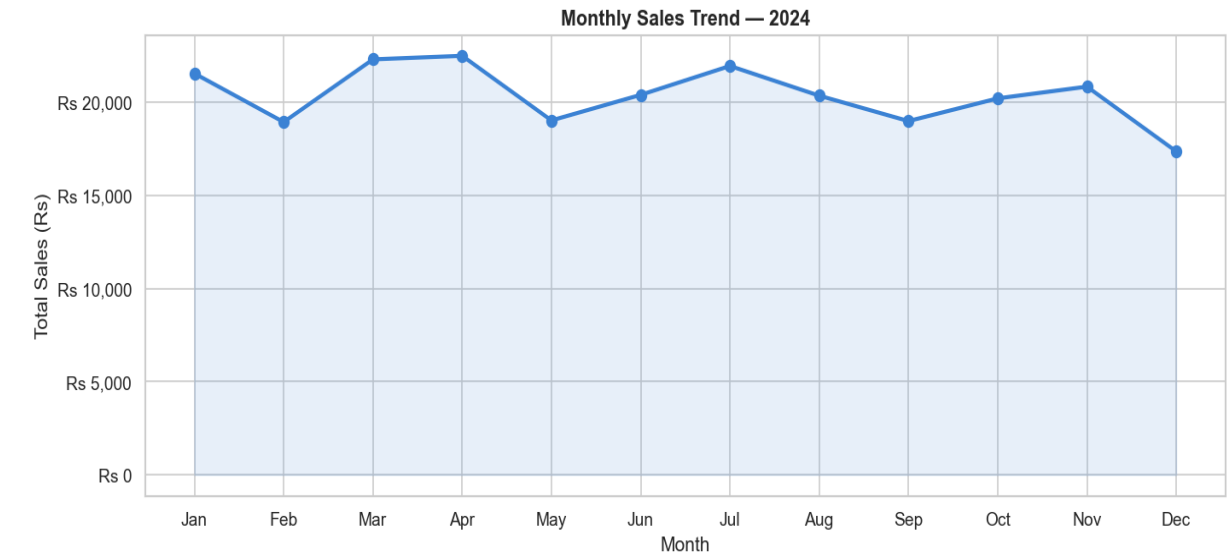


Figure 8: Monthly Sales Trend (2024) — Peak: April 2024 (Rs 22,495.00) | Lowest: December 2024 (Rs 17,361.50)

13. Rating Analysis

The average customer satisfaction rating is 3.99/5 across all 500 transactions. No transaction received a rating below 3.5, establishing a solid minimum floor. Rating ranges from 3.5 (minimum) to 4.8 (maximum). Approximately 17% of transactions received excellent ratings (4.5–4.8). Top-rated category: Dairy (avg 4.018). Branch B (Bangalore) leads with avg 4.104; Branch A (Delhi): 4.059; Branch C (Mumbai): 3.825.

Category	Avg Rating (/ 5)
Dairy	4.018
Snacks	4.011
Bakery	3.993
Beverages	3.986
Grains	3.966
Grocery	3.947

Branch	City	Avg Rating (/ 5)
Branch B	Bangalore	4.104
Branch A	Delhi	4.059
Branch C	Mumbai	3.825



Figure 9: Customer Rating Distribution (2024)

14. Business Decisions

• **Decision 1 — Protect Premium Product Supply Chain**

Cheese (Rs 58,066.08 — top revenue product) and Paneer (Rs 200/unit — highest unit price) together account for a disproportionate share of total revenue. Establish minimum stock thresholds and automated reorder points at all three branches.

• **Decision 2 — Launch Q4 Revival Campaign**

Q4 (Oct–Dec) generated Rs 58,409.00 across 123 transactions — 23.9% of annual revenue, the weakest quarter. December 2024 is the lowest-revenue month (Rs 17,361.50). Design a Diwali/year-end promotional calendar. Target: lift Q4 revenue to Q1 level (Rs 62789).

• **Decision 3 — Convert Normal Customers to Members**

Normal customers (236 txn, 47.2%, ATV Rs 494.98) are closely matched with Members (264 txn, ATV Rs 483.32). A 'Join at Checkout' programme with same-day UPI cashback can grow Member share and retention.

- **Decision 4 — Invest in UPI Infrastructure Reliability**

UPI accounts for 194 transactions (38.8%). Combined cashless share: 70.4%. Any UPI downtime directly impacts the largest payment channel. Ensure dedicated connectivity with backup at all POS terminals.

- **Decision 5 — Branch-Specific Promotional Strategies**

Mumbai (Rs 87,956.58): premium product spotlights. Bangalore (Rs 78,555.00): health product expansion + Coffee bundles. Delhi (Rs 77,899.50): Member loyalty double-points months.

- **Decision 6 — Expand Bakery and Grocery Range**

Bakery (Rs 11,000.00, 40 txn) and Grocery (Rs 8,500.00, 34 txn) are single-product categories. Adding 2–3 products each could grow their combined share from 9% to 15%+ of transactions.

- **Decision 7 — Target 4.2+ Average Rating**

Current average: 3.99/5. Reduce checkout wait time, ensure premium product display quality, train staff on customer engagement. Target: 4.2+ average within two quarters.

- **Decision 8 — Reduce Cash Dependency at Branch A (Delhi)**

Branch A has the highest cash transaction ratio. Run a 'Go Digital' UPI incentive campaign; track monthly cash vs digital ratio at Branch A.

15. Conclusion

This project analysed 500 supermarket transactions (January 2024 – December 2024) across three branches in Delhi, Bangalore, and Mumbai, using Python (Pandas, Matplotlib, Seaborn) in Google Colab. All results verified from data/supermarket_sales.csv.

Total revenue: Rs 244,411.08 | ATV: Rs 488.82 | Average rating: 3.99/5 | Total quantity: 2,768 units.

Peak revenue month: April 2024 (Rs 22,495.00). Lowest revenue month: December 2024 (Rs 17,361.50). Q4 is the weakest quarter (Rs 58,409.00 — 23.9% of annual revenue). Cashless transactions: 70.4% (UPI 38.8% + Credit Card 31.6%). Top product: Cheese (Rs 58,066.08). Top category: Beverages (Rs 117,511.08 — 48.1% of revenue).

Eight targeted business decisions — from supply chain protection to Q4 revival campaigns and customer loyalty conversion — provide a clear, data-backed roadmap for the next operating year.

Verified KPIs (from data/supermarket_sales.csv): Total Sales Rs 244,411.08 | Transactions 500 | ATV Rs 488.82 | Rating 3.99/5 | Qty 2,768 | Top Product: Cheese | Top Category: Beverages | Top Branch: Branch C | Top Payment: UPI | Peak Month: April 2024 (Rs 22,495.00) | Lowest Month: December 2024 (Rs 17,361.50)